

Name: _____

Due: October 9, 5PM via e-mail or Edwards Mailbox

1. ^{131}I is used for targeted treatment of thyroid cancer, since these tumors are partial to absorbing iodine and low-energy betas from the decay tend to stop in the cancerous tissue. To create ^{131}I , ^{131}Te is created in large quantities in a very short time by irradiating ^{130}Te in a high neutron-flux reactor. How many hours after you have created the ^{131}Te sample will you have the maximum amount of ^{131}I ?

2. Calculate the half-lives for α decay from ^{235}U , ^{231}Pa , ^{227}Ac , ^{223}Fr , and ^{219}At , from the ^{235}U decay chain. Rank the half-lives and compare to the experimental ranking. Show your calculation for one case but use a spreadsheet or program for the others.

3. ^{196}Au can decay by β^- , β^+ , and EC. Find the Q-values for the three decay modes.

4. Classify the following decays according to allowed/super-allowed/forbidden. Indicate for the allowed/super-allowed cases if they are Fermi or Gamow-Teller Transitions.
 - a. $^3\text{H}(\frac{1}{2}^+) \rightarrow ^3\text{He}(\frac{1}{2}^+) + e^- + \bar{\nu}_e$
 - b. $^{26}\text{Al}(5^+) \rightarrow ^{26}\text{Mg}(0^+) + e^+ + \nu_e$
 - c. $^{26m}\text{Al}(0^+) \rightarrow ^{26}\text{Mg}(0^+) + e^+ + \nu_e$
 - d. $^{14}\text{C}(0^+) \rightarrow ^{14}\text{N}(1^+) + e^- + \bar{\nu}_e$

5. The National Nuclear Data Center (NNDC) provides a wealth of information relevant to nuclear physics, in the form of online databases, and interactive calculators, for example. Specifically, tools exist to calculate $\log(ft)$ values from experimental decay data. Take a look at this tool at www.nndc.bnl.gov/logft. Discuss the input parameters to this calculator in terms of your knowledge of the expression for the ft value in β decay. Comment on any information you believe is missing (assumptions that must be made behind the scenes of the calculation), or any input values you do not believe are strictly required.

 Try using the calculator yourself. Assume that you have made the measurement and obtained the data summarized in Fig. 6 of [Mantica *et al.*, Phys. Rev. C 77, 014313 \(2008\)](#). Attempt to reproduce the $\log(ft)$ values in this figure. Itemize the inputs you use for the calculation. Comment on the results of your calculation, and on the magnitude of the $\log(ft)$ values in terms of the classifications in β decay (i.e. allowed, etc.)

6. Based on selection rules, which γ -decay types could link the $\frac{3}{2}^+$ state at $\sim 4\text{MeV}$ in ^{19}Ne to the $\frac{1}{2}^+$ ground state? Calculate the Weisskopf estimates for $t_{1/2}$ for these transitions and compare to the NNDC.

7. Which nucleus would generally have higher kinetic-energy fission fragments, ^{252}Cf or ^{235}U ? What is the ratio of TKE? Keep in mind shell effects on the fragment mass distribution.

8. How would you “un-grade” yourself on this assignment? What feedback would you give yourself upon completing this homework?